

Training Technique and Skill

Despite the unattainable goal of perfection, it's in striving for perfection that you depart on a trajectory toward true mastery.

Moving over stone is the essence of climbing, and therefore no subject can be more central to improving your climbing than this chapter on training skill and developing technique. Despite this, the subject of strength training tends to get all the attention and hype in conversations among climbers, and getting stronger is the most popular topic of magazine articles about climbing performance. In this book mental training and improving skill and strategy come before the subject of strength training because I feel most climbers can benefit greatly from focused training in these areas. If you want to put yourself on the fast track to the higher grades, strive to understand and apply the information contained in this chapter as much as or more than any other chapter in the book. This *will* make you a better climber.

Ironically, the majority of climbers spend little, if any, time on dedicated practice of the vast spectrum of techniques inherent to our sport. With little coaching or guidance available, most climbers unknowingly service their lust for ticking routes by constantly climbing for performance. Of course, becoming proficient (or excellent) at any sport requires focused practice of new skills and work on weaknesses that need improvement. Still, some climbers just don't want to spend time practicing on routes below their maximum grade; others avoid routes that might highlight their weaknesses. Can

you imagine a baseball player who never took batting or fielding practice outside of the competition of a nine-inning game, or a quarterback who never threw a pass except on game day?

In this chapter you will learn the three stages of motor learning involved in acquiring a new climbing skill or move, as well as how the brain creates motor programs to execute specific skilled movements and generates “software” that can approximate solutions, on the fly, to unknown moves. This is powerful information if you understand it, because it will empower you to practice more effectively than the mass of climbers, rapidly develop superior technique, and grow to become a true master of rock.

Next I present a primer on the fundamental techniques of efficient climbing movement. Beginning climbers would be wise to read the unabridged coverage of climbing technique presented in my book *Learning to Climb Indoors*. Intermediate and advanced climbers can likewise benefit by reviewing—and, if needed, shoring up their use of—the key techniques detailed in this chapter.

The action portion of this chapter is a series of practice drills that you can use year-round whether you climb indoors or out. Regardless of your tenure in the sport, significant gains in your climbing technique and overall ability will result from a commitment to regularly employing some of these drills in your routine. Apply these drills to accelerate learning (or to unlearn bad habits of movement) of the fundamental techniques outlined in this chapter.

An Overview of Motor Learning and Effective Skill Practice

The importance of motor learning theory to the subject of training for climbing became apparent to me in the early 1990s as a result of conversations with Dr. Mark Robinson and thanks to ongoing input from my wife, Lisa. As an LPGA golf pro (and climber) with an education in the field of kinesiology, Lisa occasionally discussed the methods used by elite golfers to learn very difficult skills (for me, impossible!). At about the same time, Mark Robinson (former ‘Gunks hardman and now an orthopedic surgeon in California) turned me on to an excellent text on the subject, *Motor Learning and Performance* by Dr. Richard Schmidt (1991, 2004). Robinson then penned a breakthrough piece on the application of motor learning and performance to climbing for my first climbing book, *Flash Training* (1994). Subsequently I’ve studied the subject in depth, and I’ve even had the good fortune to engage Dr. Schmidt (the founder of schema theory) and discuss motor learning and performance (MLP) and its application to climbing. The result is a much-expanded section on MLP in this edition, along with numerous powerful practice strategies to hone your skills and technique.

Types of Skills and Transference of Skill

Skill is defined as the capability to bring about an end result with maximum certainty and minimum time and energy. Climbing skills possess motor and cognitive components. Another important distinction is that between *discrete* and *serial* skills. A discrete climbing skill is a single movement with a definite beginning and end—a mantle, lunge, high step, down pull, lieback, deadpoint, and the like. String many discrete skills together, however, and you now have a more complex skilled action called a serial skill. Like a gymnast performing a routine, a climber must successfully execute specific moves, but also possess the skill to link all the moves into

a complete ascent. This explains why, in preparation for a redpoint ascent, you can’t just practice the individual moves—it’s equally important to practice connecting the moves, since this is a new skill in and of itself.

Also relevant to the process of learning and application of motor skills is the matter of skill transference, which relates to how practice of a skill in one activity carries over to enhance performance of another, different activity. Research and anecdotal evidence both confirm that transference is either small or completely absent, even between seemingly similar activities. The complexity, coordination, and integration of skilled movement are so specific that they derive very little help from other skilled movements. Therefore, practice at crack climbing will improve crack climbing skill but not face climbing ability. Furthermore, engaging in any type of “cross-training” activity will also fail to improve climbing skill. Kicking a Hacky Sack, snowboarding, slacklining, or what have you is a waste of time for the purpose of improving climbing skill (although these activities may develop cognitive attributes, such as focus, that may translate to climbing).

Sources of Sensory Information

Inherent to skilled performance is the need to receive and understand sensory information from the environment and your body. In climbing, the amount of information to process moment by moment is remarkably vast. Most obvious is the data collected through our senses regarding the climb and the surrounding environment. More subtle, but exceedingly important, is the wealth of information provided by our body relating to balance, muscle function, and quality of movement, among other things. Let’s examine more closely the two sources of sensory information.

EXTEROCEPTION

Sensory information from outside the body is called exteroception. Of the five traditional senses, vision

is the primary source of exteroception for physical activities such as climbing. Not only does vision provide the data needed to discover holds and unlock sequences, but it also offers a wealth of less conscious input on the spatial and temporal aspects of our movements.

Less obvious, but still valuable, is exteroception via touch and hearing. Whether you are aware of it or not, your sense of touch provides essential data on rock texture and quality of contact with the hands and feet. Every moment that you are on a climb, either your conscious or preconscious mind monitors this tactile input, enabling you to make subtle adjustments of grip and foot position. The auditory sense provides other novel environmental data, such as warning of approaching rockfall and confirming that a carabiner gate has closed (hearing that important *snap* of the gate), among other things.

PROPRIOCEPTION

Proprioception is your internal sense of body position and movement in space (also called kinesthesia). No matter what you do physically, proprioception provides the brain with a high bandwidth of sensory data from the nerves in all of your muscles, tendons, and joints, as well as from the vestibular apparatus of your inner ear (which allows you to sense orientation with regard to gravity). This vast amount of sensory feedback from the limbs and inner ear is processed unconsciously in doing simple tasks such as walking, cranking pull-ups, or dancing up an easy climb that requires little thought. More complex tasks, however, require conscious attention to proprioception—and it's the awareness and diligent use of this information that separate master climbers from the mass of climbers.

Awareness of specific aspects of proprioception, or what I call *proprioceptive cues*, varies on a continuum from extremely coarse and general on one end to exquisitely subtle and well defined on the other. Beginning climbers initially possess a coarse, limited sense of internal feeling as they climb. For

example, they may sense the basic quality of a foot placement, whether they are in balance, and, most obvious, how pumped they are getting! This most basic proprioception is important, but it represents just a tiny fraction of the broad bandwidth of proprioceptive cues that an elite climber can perceive and leverage.

With increasing experience (hundreds of hours of climbing) and a determination to grow your awareness of proprioceptive cues, you will come to recognize a steady stream of valuable movement cues from your body's internal sense organs. When practicing a new skill or working a move on a hard boulder problem or project climb, it is highly instructive to ask yourself, *How does it feel when I do it the right way (most efficiently) compared with when I do it the wrong way?* Making this distinction empowers you to detect flawed execution and make corrective adjustments on the fly. This subtle but immensely important skill is one of a handful of master skills that you must develop if you want to pursue your ultimate potential in climbing.

Becoming an intermediate or advanced climber, then, will correlate to your deepening sense of proprioception in a wide variety of climbing situations. Each type of rock, cliff angle, type of climbing, body position, and family of moves provides unique proprioceptive feedback that you must learn to interpret in order to move with fluidity and high efficiency. Much of this proprioception (and the resultant physical adjustments) occurs preconsciously when you are climbing submaximal sequences. Crux movements and many novel moves, however, demand full attention to proprioception, thus leaving little remaining cognitive focus for other purposes. Many falls off crux moves that you have rehearsed and seemingly wired—or off easier moves when on-sighting—are the result of poor attention to proprioceptive cues.

The bottom line: The more subtle the level of proprioception that you can perceive, the better you will be able to climb given your current skill level

and physical abilities. Serious climbers, therefore, are serious about developing their awareness and use of proprioceptive cues.

Three Stages of Motor Learning

Motor learning is the process by which we acquire physical skills. Regardless of the skill—walking, driving, climbing—learning occurs in three identifiable and overlapping stages: the cognitive, motor, and autonomous stages.

COGNITIVE STAGE

The cognitive stage of learning involves a lot of thinking (hence the name) about technique and moves, along with repeated trial-and-error attempts to execute novel moves. This first stage of learning is more visual than feel-oriented, so it helps to be shown a move and then verbally guided through execution. Early attempts are clumsy, inefficient, and jerky, expending energy and strength in wasteful ways. This is the normal *modus operandi* during your first few weeks or months as a climber, as well as when you are more experienced and make the first few attempts at a new type of climbing (crack, slab, pocketed face, etc.) or on a route that's especially hard for you.

In preparing for a climb, you tend to examine the route from the ground in an effort to figure out the moves and rest positions, and then you attempt the climb via top rope or, perhaps, leading bolt-to-bolt. The results of such early attempts typically are rough and imperfect. With continued practice, however, the quality of performance improves as you learn the feel of the moves via sharpened proprioception and knowledge of the climb.

Given the many and complex types of climbing, it may take weeks or months to advance through the cognitive stage when learning a new climbing skill set (e.g., crack climbing). Two keys to effective learning are letting go of judgments relating to poor performance (view errors as clues for improvement), and not rushing or overwhelming yourself

by trying many different types of climbing during initial sessions. Most important, strive to develop good technique by repeating moves (and entire routes for that matter) until you can climb them smoothly and efficiently. Remember, success by way of thrashing up a climb is really a failure when it comes to learning—it wires bad habits of movement into the brain, thus developing bad technique rather than good technique!

It's during this first stage of learning that a climbing coach is most valuable. Not only can the coach demonstrate new skills and proper ways of movement, but she can also provide verbal cues to guide proper posture, body positioning, tactics, and problem solving. The coach's feedback should be positive and highly specific, yet not overwhelming in volume—the focus should be on correcting major flaws, not tweaking minor technical details. Finally, it's only in this stage of learning that the coach (or others) should provide significant beta, or sequence instruction. A beginner generally lacks the experience to see some of the necessary holds and to read complex sequences, so providing beta is actually a good way of growing experience and creativity in sequencing.

MOTOR STAGE

The motor, or associative, stage is less a product of self-conscious effort and thought than one of automatic increases in the efficiency and organization of the activity by the nervous system and brain as a response to continued practice. The brain wires new neural associations as a result of regular practice of a class of skills and multiple attempts at a specific movement. The climber exhibits more fluid, confident, and economical movement. Energy expenditure decreases, and the natural momentum of the body and limbs is used to advantage. This marked increase in economy of movement is the hallmark of the motor stage of learning.

When working a climb, this stage is represented by the attempts at redpoint when the moves and

clips are known, and the goals are to develop efficiency and conserve power and endurance for the cruxes. The underlying factors involved in this stage differ from those that lead to early success. Here they involve improving proprioception, the accuracy of limb movement, the speed of detection and correction of minor errors, and the sensitivity of the performance to anxiety, doubt, and so on. These things are obviously less available to conscious awareness or control and are thus acquired only through dedicated practice and the chase of perfection.

In this stage the goal of action becomes more refined and demanding. The moves must be done efficiently with strength to spare, not eked out in desperation. Early, crude success should not be accepted as “good enough,” since this will not lead to the best ultimate development of technique and efficiency. Having demanding goals has been shown experimentally to produce both better performance and faster gains. The goal should be to perfect movement and dominate at a certain grade, not just get by at it.

Coaching input during this stage should be far less than during the cognitive stage. The climber must be allowed to self-diagnose errors and make a first attempt at correction. In many cases the coach may want to save feedback until after the climb—then, in a relaxed setting, talk through specific errors and techniques to practice in the future. Two important keys to rapid learning during the motor stage are broadening the skill set through exposure to new types of moves and climbs, and extensive use of proprioceptive cues to increase awareness and feel of technically correct movement. The climber should be encouraged to talk through (and even write down) what he feels when performing specific moves, although the coach may need to draw out this information by asking, “How does the move feel when you do it correctly versus when you do it poorly?”

AUTONOMOUS STAGE

The final stage of learning is called the autonomous stage. At this point many climbing moves are

automatic and require very little conscious attention, because movement has reached a stable and polished form. You can often do other things while in this state: For instance, you can carry on a conversation while driving a car, or you can decipher and send a moderate route in perfect form, on-sight. This apparent ease of execution is one aspect of the flow state, or “zone,” so often touted by elite athletes. In climbing, it is reached only through dedicated, disciplined, long-term practice.

Veteran climbers with many years or decades of experience exemplify autonomous performance when they fluidly on-sight the majority of nonmaximal climbs they touch. The average climber with just a few years’ experience may have this level of automatic, efficient movement on a well-rehearsed redpoint ascent or when on-sight climbing well below maximum ability. Given the complexity of climbing, it usually takes more than ten years to reach a consistent autonomous stage of mastery. Still, such elite climbers always have room for learning and refining skills, and the very best embrace a lifelong, ever-learning mind-set to exploring their potential.

As the name implies, climbers in this performance stage need little input from a coach—the fundamentals are well known, and the climber is capable of self-correcting errors and self-directing practices and training. Coaching input here should focus on questioning the climber about how he feels, with the goal of enhancing his awareness of lingering (often very subtle) technical and mental constraints on performance, and helping develop an effective long-term training strategy.

Motor Programs and How We Perform Novel Moves

Despite being as intuitive and natural as walking or running, climbing can be a remarkably complex and demanding activity. Consider that the climbing gyms and crags of the world offer a playing field of infinite variation and demand for skilled

performance. Compound this with the potential for adrenaline-releasing risk and the perplexing challenge of ascending a gigantic wall, and it becomes apparent that climbing is indeed a most complex sporting activity.

The goal of this section, then, is to provide a primer on the theory of skilled performance that empowers you to learn and develop climbing skills most effectively and rapidly. While the mass of climbers stumble through the maze of trial-and-error learning, your knowledge of how the brain wires motor programs and executes novel moves is a lever that opens the door to techniques for accelerating technical improvement (described in the next section).

MOTOR PROGRAMS

Discrete climbing skills are directed by motor programs wired into long-term memory. These motor programs, which define and shape movement, become more stable, elaborate, and long lasting as you progress through the three stages of motor learning described earlier. With consistent, quality practice, the programs become highly precise and largely unconscious, thus freeing attention for other matters such as finding the next handhold or remembering the sequence. Conversely, climbing skills that you rarely practice (or avoid) will be represented by less detailed, unstable motor programs that may lead to poor execution and require high attentional demands.

A highly skilled climber can often look at a section of rock, see the moves, and know what skills he will need to employ. For example, he might view a short boulder problem and see a lieback move that leads to a backstep move, followed by a two-foot deadpoint move to a good hold. In ascending the boulder problem, motor programs for “lieback,” “backstep,” and “deadpoint” yield a sort of movement script directing essential details of movement, like the muscles to be used and in what order, the force and duration of each contraction, and such.

Still, the motor programs will not specify every aspect of movement, and so the climber will make many tiny reflexive, preconscious, or conscious proprioceptive adjustments that modify the commands of the movement script.

A less skilled climber may be challenged in many ways by the same boulder problem. First, she may lack the experience and cognitive skills to “see” the necessary moves. Furthermore, she may possess less refined motor programs for the lieback, backstep, and deadpoint skills that are necessary to ascend the sequence. Chances are she will need numerous attempts to feel out the moves and gather proprioceptive feedback on body position, the muscles to be used, and the force of contraction, among other things. Repeated blocked practice attempts may eventually lead to a successful ascent of the boulder problem.

PERFORMING NOVEL SKILLS

The hallmark of an expert climber is the ability to on-sight climb at a high level on a wide variety of terrain and rock types. But how does this climber execute novel moves with a high rate of success? After all, our playing field has infinite variability—so even a well-traveled professional climber will fail to experience every possible move and body position. Understanding how novel movements are generated will empower you to practice more effectively so that you can become a master at on-sight climbing through moves you’ve never before experienced!

Let’s use the boulder problem example above, which concluded with a deadpoint to a good hold exactly 24 inches away. Given the infinite variability of climbing moves and rock surface, it is unlikely that the climber will have previously performed a deadpoint move, from a backstep position, to a hold exactly 24 inches away. However, if the climber has thrown deadpoint moves from several different distances (say, from 15, 21, 28, and 30 inches away) and from different body positions and rock angles,

he will likely be able to execute the novel move on-sight. This is because all the different deadpoint moves call the same “deadpoint” motor program into use, and this motor program is scaled to fit the novel situation according to a learned set of rules, called a *schema*.

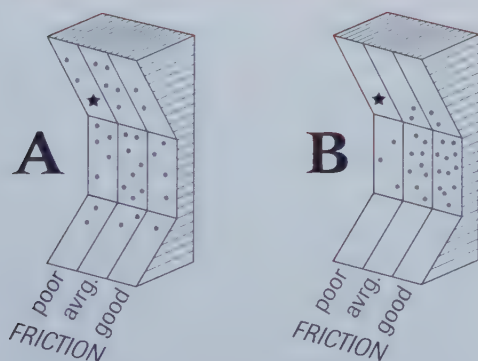
A schema is a set of rules, developed and applied unconsciously by the central nervous system, that enables you to adjust a motor program for different environmental conditions (hold location, rock angle, friction properties, and such) by changing parameters of muscle force, body position, and speed of movement (see figure 4.1). Becoming a proficient on-sight climber, therefore, isn’t just a matter of learning all the different classes of skills (jamming, side pulling, down pulling, lunging, flagging, etc.), but it also demands that you practice these skills in a wide range of configurations and settings. Such variable practice (more on this in a bit) refines the

existing schema–rule for each generalized motor program, thus allowing more accurate estimation of the necessary parameters for execution of a skill in a novel situation.

The practical application of schema theory should now be obvious: Upon learning a new climbing skill, say finger jamming, you’ll want to practice finger jamming in cracks of different sizes, on different wall angles, and on rock with different frictional properties. Doing so will expand your use of the finger-jamming schema–rule to effectively ascend finger cracks at almost any crag on the planet. The same goes for other climbing skills—strive to expand their use to a variety of rock types and terrain, and you will be on your way to becoming a master of rock!

Conversely, if you climb at only a few cliffs and favor a specific type of climbing, you will develop fewer motor programs and less refined schema–rules for each. These motor programs, no matter how well learned, will work only for similar situations—and they may not apply particularly well at the outer limits of difficulty at these crags. Worse yet, when you travel to new areas, your limited skills and schema–rules will leave you climbing at a much lower grade or flailing on routes of the grade you’re accustomed to sending at your home area.

Figure 4.1 Developing Schemas



Climber A has practiced a specific skill over a wide range of conditions (dots) and thus possesses a more refined schema-rule. She will be able to closely approximate the novel move (star)—and possibly on-sight it!
Climber B has practiced the skill primarily on vertical rock with good frictional properties. Therefore, he is likely to struggle in executing the novel move (star) on slick, overhanging rock.

The Importance of Feedback in Developing Expert-Level Skill

In wrapping up our study of motor learning, I must stress the importance of growing your awareness of feedback in all your climbing endeavors. No matter the task, you must consciously compare the ongoing feedback with the desired goal to determine the nature and amount of error. You are then empowered to amend your actions in a way that improves efficacy and moves you closer to your goal.

The desired goal in learning to climb should be to expand your skill set, improve quality of movement, and practice in ways that will accelerate your journey toward becoming an advanced climber

steeped in autonomous-level performance. Making the most of both extrinsic and intrinsic feedback will hasten progress up the learning curve. Let's examine these two critical types of feedback.

EXTRINSIC FEEDBACK

Extrinsic feedback is information provided by some outside source, such as a belayer, coach, or videotape, that augments your internal sensory feedback. Extrinsic feedback is an inherent part of cognitive-stage learning, in which a beginner receives extensive verbal instruction to guide learning of new skills and proper movement. As you progress to intermediate-level ability (the motor stage), it's important to accept—and ask for—a lesser amount of extrinsic feedback. Growing your abilities demands that you learn to utilize the high bandwidth of internal feedback (more on this in a bit) in place of extrinsic feedback in self-analyzing and self-correcting movement error. Advanced climbers need very little extrinsic feedback in daily climbing and training activities, although the objective feedback of a coach is valuable in the case of a performance slump or long-term plateau in improvement.

A most common form of extrinsic feedback is *beta*—highly specific instruction on the proper sequence for a given climb. Beta is extremely helpful in developing hold recognition and sequencing skills during the formative stage of learning. As you progress beyond the cognitive performance stage, however, continued use of beta will handicap the development of problem-solving skills and other advanced cognitive abilities. What's more, long-term reliance on beta will lead to a false sense of ability, in that your climbing performance will degrade significantly when beta is not available. The bottom line: Beyond your first few months as a climber, make it a practice to shut off beta from your partner (or others); the exception is those rare cases when beta is essential for safety or speed.

INTRINSIC FEEDBACK

Intrinsic feedback is sensory information via your five senses and proprioception gleaned from your muscles, tendons, joints, and the vestibular apparatus of the inner ear. The most obvious intrinsic feedback is what you visually observe as the gross results of your efforts: falling off a move, seeing that a hold is not what you expected it to be, or discovering that a sequence you visualized won't work. Most climbers, regardless of ability, have good awareness of such basic intrinsic feedback.

It's the subtler intrinsic feedback of "feel" (proprioception) that beginners—and many non-beginners as well—lack awareness of. It's understandable, of course, for a novice to lack the subtle senses of touch, body position, and levels of muscular tension, for example, due to the aggregate overwhelm of fearful thoughts, technical struggles, and getting pumped! As your quality of movement improves (the motor stage) and you gain greater control over fear and excessive muscular tension, the subtler proprioceptive cues begin to reveal themselves. Still, you need to direct your attention inward to fully harvest the proprioceptive cues and come to understand and leverage them completely. Growing your awareness and use of proprioceptive cues is a mental skill that you need to develop through disciplined effort.

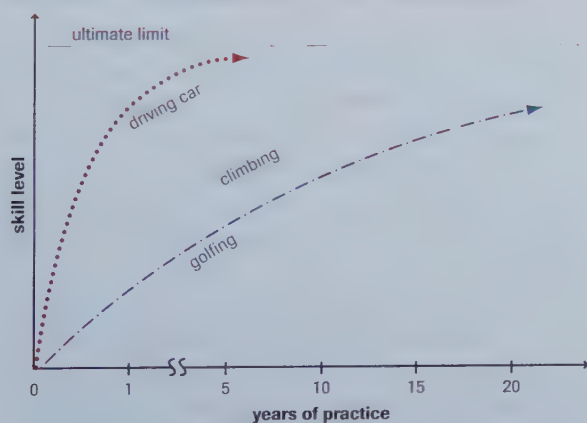
As described earlier in this chapter, it's a beneficial practice method to talk through and write down proprioceptive cues that you discover for specific types of moves and crux sequences of project routes—this process will foster uncommon sensory awareness, accelerate motor learning, and put you on the fast track to the higher grades. For many advanced climbers, use of proprioceptive cues will become largely preconscious; the exception may be in stressful, effortful moments when movement quality suffers and you need to proactively modulate tension and consciously utilize proprioceptive cues to maintain high-quality movement.

Rate of Improvement and Your Ultimate Skill Level

According to the Law of Practice, performance improves rapidly from its baseline level when the activity is first practiced and continues to improve in gradually decreasing amounts as ability approaches some ultimate (personal) skill level (see figure 4.2). In learning a simple task, like driving a car, it only takes a few weeks to become fairly skilled. Beyond this, all the thousands of hours you spend driving over the rest of your life will yield only a small amount of improvement.

The learning curve for complex activities, like golf or climbing, also rises rapidly as a result of the initial practice sessions. Because of the high complexity and wide range of skills inherent to these sports, however, you can continue to improve for many years, even decades. In fact, golf would seem to be the more technically difficult sport of the two, since several gifted climbers have reached world-class status in less than five years, while the best golfer ever, Tiger Woods, took fifteen years to ascend to that level—and that was unusually fast!

Figure 4.2 Rate of Skill Acquisition



Performance improves rapidly from its baseline level when the activity is first practiced and continues to improve at a slower rate as ability approaches some ultimate personal skill level.

As discussed in chapter 1, you can maximize your rate of improvement in a specific style of climbing by focusing your practice and training in that one area. It's increasingly common to observe climbers who quickly progress to a high level of ability in one type of climbing—say, gymnastic bouldering or overhanging crimpfest routes. Since both of these endeavors require only a small subset of total climbing skills, it's possible to focus your practice on developing the small set of motor programs necessary to excel. The trade-off is that your limited motor programs will be a handicap when attempting routes outside your area of specialization—these climbs will feel hard for the grade, perhaps even “impossible.”

The bottom line: Becoming a proficient, all-around climber is a long-term proposition. The most intelligent approach to training for climbing would be to emphasize learning skills over getting wickedly strong during your first few years in the sport. Developing refined schema-rules for the many different climbing skills will take many years of climbing at a wide range of areas and rock types. So while you may become a highly skilled specialist climber in just a few years, you can continue to expand the depth and breadth of your skills even after twenty or thirty years (I know that I still am!).

Practice Strategies to Accelerate Learning of Skills and Enhance Schema-Rules

If you want to maximize improvement, then you need to know how to practice optimally. Acquiring a new skill requires a progression through the three stages of motor learning before you will be able to use the new skill efficiently and intuitively. Depending on the difficulty of the skill, this process may be slow, awkward, and frustrating. Therefore, discipline and an intense belief that you will become excellent in the skill (with effort) are fundamental to the process. You must not try to get around having to learn it by relying on your strengths in other areas.

Unfortunately, many climbers do exactly this when they lunge, swing, or scum wildly through a sequence that could be done more efficiently by learning a new move, technique, or body position. Another example plays out when a competent gym or face climber first attempts to learn crack climbing. The common tendency is to avoid the awkward—and possibly painful—jams and paw up the face on either side of the crack. In this case the climber will rapidly fatigue and likely exclaim that the route felt way harder than its grade (which of course it did, since the climber wasn't using the proper technique for ascending the route).

The moral of the story is that you must convince yourself that no matter how hard a new move feels at first, it will become easier to execute—and perhaps even intuitive—as the result of intelligent practice. Toward this end, keep in mind that you will become most competent at a skill by practicing it on as many different rock varieties, angles, and settings as possible.

Another important principle states that learning new skills is difficult in states of fatigue, stress, fear, and urgency, whereas a fresh, relaxed, and confident approach yields rapid acquisition of new skills. Consequently, it's best to practice new skills early in your training sessions and to employ liberal hangdogging for blocked practice of crux moves. Interestingly, you can solidify skills you already possess by practicing them in a fatigued state! If all this motor learning theory is beginning to overwhelm you, worry not—following are five powerful practice strategies that distill and apply this information to produce optimal skill-learning results.

MODELING ADVANCED CLIMBERS

Modeling is a powerful technique for learning basic skills and climbing strategy. It's best used in a climbing gym where you can observe the movements, positions, techniques, and tactics of a more advanced climber, and then immediately give them a try on your own. Make a mental picture of what

you want to attempt, and use that vision as a starting point. Experiment, modify, and make the move your own. In practicing the new move, progress through the following four practice strategies: blocked practice, variable practice, fatigued practice, and random skill practice (described below).

You can also model what you observe at the crags. In addition to actual moves, take special note of the tactics and strategy used by high-end climbers. For example, how do they work crux sequences? Where do they find rests? At what pace do they climb? How do they go about equipping routes? Again, it's best to first experiment with your observations in the gym before testing them outdoors.

Although modeling technical skills is a powerful weapon for your arsenal, copying an elite climber's strength and power training program is usually a big mistake. Remember that elite climbers have spent years conditioning their muscles and tendons to withstand extreme levels of stress. To train as they do without this long-term preparation could be disastrous.

BLOCKED PRACTICE

Blocked practice—identical repetitions of a specific move—is the most popular method of practicing a hard climbing move because it produces rapid learning of the skill in that specific situation. In learning the undercling move, for example, you would repeat the same undercling move over and over in order to refine your body position and the application of force needed to optimally perform the move.

Upon development of “feel” and early success at a new skill, however, a radical change is needed. Further blocked practice will have little value and may even result in a false sense of confidence and poor use of the skill in novel settings. Returning to our example, suppose you only practiced the one basic undercling move you first learned at the gym. Despite your expertise at that specific undercling

move, you will struggle and likely fail on underclinging moves on different wall angles and on the infinite playing field of outdoor climbing. The same phenomenon is seen in other sports, like a golfer who hits great wedge shots from the perfect lie of a practice range mat (blocked practice), yet on the golf course is unable to hit a solid shot from the novel lies of the fairway and rough. These examples show that beyond the first few successful trials of a new skill, blocked practice is for blockheads! Upon achieving initial success at a new skill, you must graduate to variable and random practice.

VARIABLE PRACTICE

As explained above, learning a new skill in a specific way is just the first step in the learning process. The ultimate goal is instant and proficient use of the skill in any new situation you come upon. The tried and proven way to do this is with variable practice.

Suppose you've just learned the drop-knee move on a vertical indoor wall with large, positive holds. To incorporate variable practice, you would now change the "route" conditions slightly and attempt the same drop-knee move again. After a few reps in this new setting, you'd again modify the route by changing the hold spacing and wall angle to further expand use of the skill—then continue this progression to the point that you could perform the drop-knee move in a variety of random settings. Such variable practice will refine schema-rules to direct effective execution of the drop-knee motor program over a wide range of conditions (angle, hold size, rock type, and frictional properties) that you may encounter in the future.

FATIGUED SKILL PRACTICE

Earlier I mentioned that you can increase your command of a skill by practicing it while in a fatigued state. In fact, beyond the initial successful trials of a skill, practice should be performed with variable conditions and levels of fatigue and never again "blocked." This may increase your rate

of failure at doing certain moves—but remember, performance isn't your goal, practice is! The benefits of this practice, no matter how poor it feels, will become evident in the future. Besides, this concept actually makes good sense. If you want the ability to stick a deadpoint in the midst of a dicey lead climb while pumped, you'd better log some deadpoints in various states of fatigue during practice.

Here's the best approach. Use the first thirty minutes of your workout (while fresh) to train new skills, then move on to chalking up some mileage on a variety of routes. After an hour or so, or when you're moderately fatigued, attempt several reps of recently acquired moves and sequences. As fatigue increases, finish up with some reps of sequences or boulder problems you have more completely mastered.

In the context of a two-hour climbing gym workout, this rule emphasizes the benefit of squeezing in a greater volume of climbing "practice" ascents over doing just a few "performance" ascents with extensive rest in between. The long rests and performance climbing may make you look better, but the greater volume of practice will make you climb better!

RANDOM SKILL PRACTICE

The ability to on-sight a sequence of novel moves on "foreign" rock is the ultimate goal of your skill practice time. Toward this end, the best workout approach is a randomized free-for-all of skill types. This highly effective method is widely used in other sports and should not be overlooked by climbers as optimal training for the unknown.

There are two approaches to random training of climbing skills. First, on an indoor wall, attempt to link a sequence of very different bouldering moves. Contrive an unusual sequence of moves that will call a wide range of skills into use, and make several attempts at sending it. Alternatively, team up with your most deranged friend for a round of the Stick Game (described later in this chapter). Take turns pointing (with a broomstick or laser pointer) each

Tips for Rapid Learning of Skills and Developing Good Technique

1. Engage in regular climbing “practice.” That is, go climbing with the intention of learning new skills and improving quality of movement, with little regard for absolute difficulty. Climb on as many different types of rock, wall angles, and areas as possible to build diverse skills and true climbing expertise.
2. Practice new skills and techniques early in the session while you are physically and mentally fresh.
3. Use blocked practice to accelerate learning during the initial trials of a new move, skill, or sequence. After two or three successful repetitions, cease blocked practice in favor of variable and randomized practice.
4. Employ variable practice to expand command of newly acquired skills over a wide range of conditions (angle, hold size, rock type, and so forth). Vary the “route” conditions more than you expect would be the case in a real-life climbing situation.
5. Practice known skills while in varying states of fatigue to increase your mastery of them and to build long-term retention.
6. Use random practice—climb a random series of moves back-to-back—to enhance proficiency of serial skill performance. Another random practice strategy is to climb several very different routes back-to-back to mandate recall of many different motor programs.
7. Model the techniques and tactics of advanced climbers. Stretch your imagination for climbing movement and strategy by observing and mimicking the techniques and tactics of top climbers.
8. Aspire to dominate at a climbing grade, and not just get by. Focus practice on routes at or just below your maximum difficulty, and resist the urge to constantly work routes beyond your ability level.
9. When working a route, resolve to find the best way to do a move or sequence, and resist the urge to just thrash up the route and deem that acceptable. As a practice method, climb a route several times to identify and learn the most effective and efficient moves and tactics.
10. Maintain a long-term perspective toward learning to climb. No matter how fast you improve or how hard you climb, realize that you can still improve technique and learn new skills—even after ten, twenty, or more years!

other through an unusual sequence of widely varied skills and movements. Don’t get too wrapped up in performance outcomes—if you link a random series of moves, then you are a winner in terms of developing superior climbing skill.

Another powerful method of random skill training is to climb a series of widely differing routes in rapid succession. A commercial gym with many different wall angles, a few cracks, and a roof or two is ideal. Team with a partner and top rope ten to fifteen routes of different character over the course of an hour or so. The first route may be a vertical face, the next a slab, the third a finger crack, the fourth

an overhanging pumpfest, the fifth a hand crack, the sixth a roof route, and so on. This rapid recall of a wide range of motor and cognitive skills is like taking skill-fortifying steroids!

Fundamental Techniques

The essence of climbing is a dance up the wall using the four points of contact as your dance steps. Since each climb possesses a novel configuration of hand- and footholds, your challenge is to unlock the perfect sequence of moves and leverage your points of contact into this dynamic dance.

In executing any physical skill—whether it’s shooting a basketball or simply running—there are fundamental techniques that represent optimal use of body position, leverage, and physical energy. While the specific techniques may be hard to observe with an untrained eye, just about any novice can spot an athlete steeped in the fundamentals: Her movements are smooth, crisp, and confident, and her demeanor reveals a poise and ease of execution, despite inherent difficulties of the situation. The bottom line: Fundamentally sound movement affords perfect economy and looks easy.

Unfortunately, becoming a highly skilled, proficient climber rarely just happens—it results from a conscious decision to develop superior technique via disciplined long-term practice. Make it your

goal to learn to climb every move and every route in the optimal way, and not to be satisfied to succeed with sloppy, inefficient movement. Excellence in climbing comes only to those with knowledge of the fundamentals and a desire to make them habit. To help you on this journey, this section provides a primer on twelve fundamental techniques that you must practice to the point of mastery. As you read through each section, ponder your current level of competency in each fundamental technique and make it an urgent part of your training-for-climbing program to improve these weak areas. For a more tangible record of your technical ability, score your command of each fundamental technique using the checklist assessment in table 4.1.

Table 4.1 Score Your Technical Ability

Evaluate yourself in each area and check the box that best represents your current ability.				
Technique	Excellent	Good	So-So	Poor
Precise, quiet foot placements that carry your weight				
Handholds gripped lightly; arms in a secondary role				
Use of the Left-Right Rule for stable movement				
Economy of movement (rhythm, pace, poise)				
Use of rest positions				
Use of non-downpull and nonpositive handholds (side pulls, underclings, slopers)				
Use of flagging to aid stability and prevent barndooring				
Hand-foot matching and mantling				
Twist lock, backstep, and efficient movement on overhanging terrain				
Use of creative footwork (heel and toe hooks, and knee locks)				
Dynamic moves (deadpoints and lunges)				
Jam crack climbing				

Precise Foot Placements That Carry Your Weight

Given that your legs are stronger than your arms, the first fundamental of climbing is that the legs should do the majority of the work. The exceptions to this rule are severely overhanging routes, which demand greater use of the arms (more on this in a bit). The process of effectively using your feet begins with spotting the footholds and positioning your feet on the best part of each hold. Directing your foot placement demands attention to detail beyond that given to hand placements. Whereas

handholds are easy to inspect, the greater eye-to-foot distance commonly leads to less-than-ideal foot placements. Furthermore, your feet don't provide the same degree of feel as the hands, making the quality of each foot placement more difficult to assess. For these reasons, developing good footwork is an attribute that you must make happen via constant foot focus and practice.

The other aspect of sound footwork is proper alignment of your center of mass directly over a foothold. Balance, stability, and application of force are optimized when your center of mass is



- 1. Keep your center of mass over your feet so they carry most of your weight. This requires a hips-out position when climbing slabs.***



- 2. On vertical rock, it's essential to keep your hips in near the rock to position your center of mass over your feet.***

positioned directly over your feet, forming a line perpendicular to level ground. On a less-than-vertical wall or slab, this requires a hip position out from the wall and over the foothold. On a near-vertical climbing surface, you simply need to keep your body position straight and over your feet as much as possible. When the climbing wall overhangs, it becomes impossible to position your weight over your feet, so a new fundamental skill takes over (see “Twist Lock and Backstep” later in this chapter).

Grip Handholds Lightly and Let Arms Play a Secondary Role

In a sport where anxiety and fear often rule, it’s understandable that many climbers hang on with their hands for dear life. This tendency manifests itself with overgripping of the handholds and unnecessary muscling of moves with the arms. The end result is rapid fatigue, pumped forearms, and an eventual need to hang on the rope in order to rest and recover.

You can avoid this outcome by practicing—and making habitual—the fundamentals of proper hand and arm use. These critical skills include gripping each handhold with the minimum force required, using the arms mainly for balance and not as a primary source of locomotion, and pushing with the feet in unison with modest arm pull.

Begin by making each hand contact a conscious process. Whereas many climbers just grab a hold with little thought and continue with the process of climbing, you must make each hand placement a thoughtful act. First, consider where the best place is to grab the hold. It’s not always on the top of the hold, and it often relates to the location of your last foot placement. As you grab the hold, focus on using a light touch that yields soft forearms. Sure, certain holds will demand that you bear down hard on them, but many don’t. Your goal must be to try to use each hold with a light touch, and then increase the gripping force only as much as is required for the move at hand. This process

of minimally gripping each handhold takes but a split second, yet it’s a master skill that separates the best from the rest. Commit to targeted practice of this important skill (use the drills detailed on page 83, and in time it will actually become an unconscious act as you cruise up hard routes with high efficiency!

Beyond gripping the rock, you need to decide just how much you need to pull down on a given handhold. As discussed earlier, it is imperative that you push with your feet and let the leg muscles carry the load. In climbing a slab or near-vertical wall, think of your arms as points of contact that simply prevent you from falling backward off the wall. In climbing a ladder, for example, your legs do all the work while the arms mainly provide balance. While rock climbing is far more complex, hold this model in your mind as the ultimate goal—the arms maintain balance while the legs drive movement.

Still, there will be occasions in which your arms will need to briefly carry much of your weight. In these situations it’s imperative that you pull quickly through the strenuous move in order to get back onto good footholds that can carry your weight. A final tip for strenuous situations is to maintain straight arms as much as possible—this way, much your weight is supported by the skeletal system of your upper body and not by your muscles.

The Left–Right Rule for Stable Movement

The magic of efficient climbing movement comes from the synergistic interaction of the arms and legs and a constant transfer of force and torque through your body. To this end, the Left–Right Rule states that maximum stability and ease of movement comes from the pairing of a left hand and right foot (or a left foot and right hand) in harmonious action. Let’s again use climbing a ladder as our model. Ascending a ladder with opposing hand–foot combinations (say, the left hand pulling and right foot pushing at the same time) is so intuitive that it’s almost impossible to climb a ladder any other way.

Suppose you tried to climb a ladder with non-opposing hand-foot combinations, such as a right hand and right foot working together; you'd immediately begin to barndoor or rotate sideways off the ladder. Thus, the Left-Right Rule is fundamental for balanced, stable movement.

While you don't need to even consider the Left-Right Rule in ascending a ladder, formulating movement up a climbing wall is much more complex: The position and shape of the hand- and footholds wreak havoc with your intuitive sense of movement. Thus, in seeking to reposition your hands and feet on the wall, it's helpful to ponder which holds will provide the best opportunity for a left-right combination. Easy climbs will often provide a pulling right hand that can combine with a pushing left foot (or vice versa). More difficult climbs tend to be more devious because the holds are smaller, farther apart, or displaced off to the side of the route line. Advanced climbers will often be able to intuit the best way to proceed; however, beginners must be willing to try difficult sequences multiple ways to discover and learn the best solution.

Move with High Economy

The technical paramount is to climb with high economy. Make these two words—*high economy*—your mantra every time you touch the rock.

High economy means discovering the way to do each move, and an entire route for that matter, with minimal energy expenditure. If you have a cat, you can observe highly economical movement firsthand. Most of the time a cat moves in a slow, smooth, deliberate way; however, sometimes a situation demands a powerful, dynamic leap to maintain perfect economy. Catlike movement should be your technical model for efficient climbing: smooth, quiet, leg-driven movements, but with an unhesitating shift to an arm-pulling, dynamic movement when it is required to most efficiently execute a difficult move. Here are five attributes of economical movement that you should aspire to acquire.

QUIET FEET

Quiet foot movements are one of the hallmarks of a climber with great technique. Conversely, feet that regularly pop off footholds or skid on the wall surface are typical of an individual possessing lackluster footwork and poor economy. For many climbers, noisy footwork is just the way they climb—it's a habit that developed over a long period of time, as well as a flaw in their technique that will prevent them from ever reaching their true potential. Your goal, of course, is to learn to climb with good foot technique even in the toughest times. This means concentrating on each foot placement, keeping your foot steady and firm to the hold, and standing up on the foot with confidence as you proceed smoothly to the next hand- or foothold.

RHYTHM AND MOMENTUM

Like any dance, climbing should have a natural rhythm that utilizes momentum and inertia. Climbing in a ladderlike motion yields the rhythm step, reach, step, reach. However, a better rhythm for effective movement is often step, step, reach, reach, since it allow the legs to direct and drive the movement. There are obviously many other rhythms, such as step, reach, step, step, reach, reach and step, step, reach, step, reach, reach. Consider that every unique sequence possesses a best rhythm of movement, and you'll eventually learn to intuit this on the fly. As a beginner climber, however, it takes a conscious effort to avoid leading with the hands in a strenuous and inefficient reach, reach, step, step rhythm. Strive to tap into the rhythm of each route, and climb accordingly.

Similarly, you want to utilize momentum in a way that helps propel successive moves. This is especially important on difficult climbs with large spacing between holds. Think of how you would move hand-over-hand across monkey bars at a playground, each movement blending with the next in a perfect continuity of motion. This style of smooth, continuous motion is critical when climbing

through crux sequences. Interestingly, many folks do just the opposite, engaging the crux sequence with measure and caution. In doing so, they not only lose upward movement but also have the inertia of stillness to overcome. Hopefully you can avoid this tendency by consciously directing a steady rhythm that maintains forward momentum through the most difficult parts of a climb.

SMOOTH MOVES AND RELAXED BODY

Smooth, fluid movement is another sign of high economy. A climber appears to flow up a route when only the muscles necessary for stability and upward motion are contracting and all other opposing, or antagonist, muscles remain relatively passive.

Unfortunately, many climbers wrought with anxiety, fear, or the need to perform find themselves contracting most all their muscles from head to toe in an attempt to hang on or fight through a crux. Such tension in the antagonist muscles forces the prime movers to work harder than needed and leads to inefficient movements and rapid energy expenditure. You can spot climbers suffering from this affliction by their stiff, rigid, and forced movements, which will likely soon have them hanging on the rope in frustration. And if they somehow succeed in fighting up the route, it will be at the expense of a great amount of energy and further ingraining of bad habits of technique.

The key to smooth, efficient movement is to maintain a high level of relaxation throughout the climb. You can best achieve this with a two-pronged approach. First, think about contracting only the muscles necessary for engaging the rock, maintaining stability, and directing movement; usually these will be the muscles of your forearms, shoulders, core, thighs, and calves. Next, switch your focus to the antagonist muscles and scan for unnecessary tension that might be developing in your upper arms, hips and legs, neck, and face. While taking deep, slow breaths, visualize this tension escaping like air from a balloon—such mental imagery

really helps the process. Now return your focus to executing the next climbing movement, but continue alternating your focus back and forth between directing movement and directing relaxation.

PACE

Pace is another aspect of climbing economy that becomes increasingly important as a route gains in steepness and difficulty. While an easy climb with large holds allows you to ascend at a leisurely pace, a crux sequence or overhanging terrain will demand that you kick into high gear and surmount the difficulty in short order. When climbing near your limit, it must be your intention to move as briskly as possible without any drop-off in technique (skidding feet, botching sequences, and such). Reduce the pace at the first sign that your technique is suffering. It helps to identify obvious rest positions ahead of time, and then make it a goal to move from one to the next as fast as possible. Ultimately, knowing just the right pace on a given route is a sense you will develop with experience. Practice climbing at different speeds and on different types of routes, and you'll quickly foster the subtle skill of proper pace.

STEADY BREATHING

Breathing is the essence of life. Steady breathing is essential to sustain the life in your climbing. Conversely, restricted breathing is akin to tightening your muscles, constraining your thinking, and fostering failure.

Obviously, a steady flow of oxygen to the muscles is important for energy production and recovery, and it's the slow, deep, steady belly breaths that best get the job done. Many climbers, however, have a tendency to shift into shallow, rapid breathing as fatigue and mental anxiety grow. Worse yet, some climbers unknowingly hold their breath at times of high stress. These are two tendencies that you must be aware of and proactively counteract if you are to climb your best.

Before every climb, take a moment to close your eyes and take several slow, deep breaths. Feel your belly expand outward as you slowly inhale, and then allow the air to escape sparingly through pursed lips in a slow ten-second count (count in your mind). Visualize a wave of relaxation washing across your body—this sets the stage for optimal performance. As you commence climbing, strive to maintain the same slow, steady breathing that you initiated on the ground. This is, of course, often difficult since a dicey sequence or strenuous move can trigger irregular breathing patterns. Consequently, it is critical that you use every rest position as an opportunity to reset your breathing cycle with a few slow, deep belly breaths. Such proactive breath control is like topping off your gas tank—do it frequently, and you’ll rarely hit empty.

Optimize Use of Rest Positions

Finding efficient rest positions is as important as finding the best way to do a crux sequence. If you miss a good rest stance, you miss an opportunity to physically recover as well as mentally “read” and prepare for the next section of the climb. Consequently, locating rest positions on a climb should be viewed with the same sense of importance as locating all the key holds.

This process begins with on-ground visualization of the route (see chapter 3) in an attempt to estimate the location and body position of rest stops. Upon reaching a rest stop, assume a body position that will allow the most fatigued muscles to rest (usually the forearms, biceps, and calves). An optimal rest position would consist of your feet in the rest-step position, legs straight, and hips over the legs or in a position midway between the feet (should they be on holds more than shoulder width apart). If the climb is less than vertical, your upper body can relax completely, and in some cases you might even be able to assume a no-hands rest position. Such a casual stance places no time limits on how long you can remain parked

there—although this luxury is rare on more difficult routes.

Rest positions on vertical to overhanging climbs make complete weighting of the feet more difficult and often impossible. While you still want to place as much weight as possible on the footholds, a significant amount of weight will remain on your arms. In this case it’s absolutely vital that you hang with straight arms, so that the bones are providing the support, not the muscles of the upper arm. It is also advantageous if you can obtain a heel hook or knee bar from which you can hang a high percentage of body weight. Still, your forearm muscles will need to contract in order to maintain a grip on the handhold. The best strategy, then, is to attain a stable stance and shake out alternate arms every five to ten seconds. This way, both arms take turns resting. At some point, however, hanging out at the rest begins to cost more energy than you can recover. It’s at this time that you need to begin climbing toward the next rest spot.

A big part of becoming a fundamentally sound climber is learning to gain brief rests in the midst of even the most difficult routes. Practice and unbridled creativity are the only two requisites for solving difficult sequences and finding vital rest positions. Remember that the best climbers are not always the strongest; their prowess instead comes from uncommon mastery of climbing economy and effective resting.

Clever Use of Opposing Forces

You have already learned the importance of the Left–Right Rule for enabling stable movement. On easy climbs this left–right combination is usually a pulling right hand along with a pushing left foot (or vice versa). More difficult climbs tend to be more devious, however, so you’ll need to consider all the other possible arm positions—side pull, Gaston, and undercling—and figure out how to match one of these with an opposing foot placement. Detailed below are a few of the most common left–right

combinations called into use on difficult climbs. You should practice each of these on the bouldering wall in order to develop its unique motor skills. Vary your hand and foot placements as much as possible to acquire a broad range of use for each hand-foot combination.

SIDE-PULL ARM AND OUTSIDE EDGE OF OPPOSITE FOOT

Side-pulling arms are a staple move on almost every moderate to advanced climb, but it's your foothold selection that often determines the effectiveness of



Pairing a side-pulling hand with the outside edge of the opposite foot (and a hip turn) provides great stability and reach.

this move. In most cases it's best to use the outside edge of the opposing foot, not the inside edge. Doing this may feel awkward at first, but you'll find a natural sense of stability once you learn to appropriately set your hips over the outside-edging foot. The key is to concentrate on rotating your hips so that the hip opposite the pulling hand is turned into the wall—that is, your face and chest will rotate toward the side pulling hand. This very stable position will allow you to step up your free (non-opposing) leg and quite possibly your free (non-opposing) hand as well. Occasionally a move will dictate that a side-pulling hand must be combined with use of the inside edge of the opposing foot. While this, too, is a fairly stable body position, it provides less reach upward with the free hand. Therefore, anytime you are struggling to reach a handhold, try using the outside edge of your shoe and a hip turn to maximize reach.

GASTON AND INSIDE EDGE OF OPPOSITE FOOT

The Gaston (aka reverse side pull) is the most unnatural and weak arm position, yet it's a fairly common move needed to unlock many crux sequences. Use of the Gaston is best opposed by the inside edge of the opposite foot. Combining a Gaston with an outside-edging foot is strenuous but doable if absolutely needed. Practice this move in a variety of ways to gain comfort and strength in its use. As with all these advanced moves, the bouldering area is the ideal proving ground to experiment with and learn the skills.

UNDERCLING AND INSIDE OR OUTSIDE EDGE OF OPPOSITE FOOT

Frequently overlooked by less-experienced climbers, undercling hand positions are often essential for unlocking difficult sequences on steep terrain. What's more, an underclinging hand helps maximize your reach with the free hand, and it positions your arm and body in a naturally strong position.



The undercling move can save the day on a reachy move lacking a positive down-pull hold. Use a hip turn and the outside edge of your opposite foot to maximize reach, stability, and power. BRUCE WILSON

Typically you will undercling a hold somewhere near your torso while you press with an opposing foot. This foot can edge with either the inside, outside, or toe portion of the shoe, although use of the outside edge is best for maximizing your reach. Remember that in edging with the outside of your foot, it's best to turn the hip opposite the pulling hand to the wall. In extreme situations you may even need to use a foot smear to oppose the underclinging hand. This is a very powerful but important move that you should train on a bouldering or System Wall.

SIDE-PULLING LEFT AND RIGHT HANDS

Use of opposing handholds is a key move for unlocking a sequence that lacks any usable down-pull or undercling handholds. Most common are two opposing side pulls that you'll draw inward to create tension through your arms, shoulders, and upper body. While you will be unable to create much upward movement, this opposition will allow you to upgrade one or both feet. Ideally, you'll want to upgrade the foot that opposes the better of the two side pulls, so that it sets up a stable left-right combination. This will enable you to release the

other side-pulling hand so as to upgrade it to the next hold. Another possibility is opposing Gaston holds. Though strenuous, you may occasionally need to grab two Gastons at or just above head height and pull outward in order to support your weight while upgrading a foot position. This is a most advanced move that requires a high level of base strength. A word of caution, however: Using a Gaston hold on an overhanging wall places great force on the shoulder joint and in rare cases can cause injury. Proceed carefully.

Use Foot Flagging to Enhance Stability

Flagging is the alternative technique for maintaining stability when a left-right hand-foot combination



Unstable moves and a tendency to barndoor can often be overcome by a flagging foot, which shifts your center of mass closer to the weight-bearing foot.

is not possible. Suppose you are attempting to use a right hand and right foot combination to propel upward movement. Upon releasing your left hand to make a reach upward, you will immediately begin to barndoor (see the photo on this page). This sideways rotation is hard to fight and often results in a fall. However, a simple flagging of the free leg (in this case the left) significantly improves stability and balance by shifting your center of mass more directly over the supporting (right) foot and under the supporting (right) hand.

Let's take a look at the two methods of flagging; which one you use depends on whether your supporting leg is edging with the inside or outside edge of the shoe. For example, consider a move in which you are edging on the inside of your right foot and pulling with your right hand. In order to avoid barndooring, you would flag the left leg across the right leg. However, if your supporting leg is edging with the outside edge of the shoe, it's best to flag the free (left) leg out left along the wall surface in an effort to find a balance point. If you're lucky, you'll find a foothold or wall feature on which to splay the flagging foot. Practice using these flagging techniques on a vertical wall, and then expand their use onto overhanging terrain.

Mantling and Hand-Foot Matching

The mantle move is often called upon to overcome a long reach between holds. Depending on the size of the hold to be mantled, you may be able to press your entire palm onto the hold or, possibly, just your fingertips. The left-right combination of pushing and pulling hands provides great stability, so you will be able to upgrade one or, possibly, both feet. In many cases you will match a foot (the same side as the mantling hand) on the same hold that you are mantling; it is then often possible to shift your center of mass over that foot and stand up.

Severe climbs frequently demand that you adroitly match your foot onto a tiny crimp or finger pocket hold. Quick, precise execution is

essential to maintain forward momentum as you progress through such tenuous sequences.

Twist Lock and Backstep on Steep Terrain

The twist lock and backstep are the bread-and-butter moves of a steep-wall connoisseur. As a climbing wall tilts back past vertical, it becomes increasingly

difficult to place a high percentage of weight on your legs. Consequently, a greater portion of body weight must be supported by the arms—which, of course, possess less absolute strength than the legs. Use of the twist lock and backstep together helps draw your body in toward the surface of the overhanging wall. This changes the force vector on the

Use of Twist Lock



1. *Without the twist lock, the climber's neutral, straight-on body position places the center of mass way out from the wall. This makes for strenuous, inefficient movement.*



2. *The twist lock with its hallmark hip turn draws the center of mass in closer to the wall, thus placing more body weight onto the feet, increasing reach and enhancing grip on handholds (thanks to the changing force vector of the arm pull).*

handholds, making them feel more positive and secure. More important, this drawing-in of the body places more weight onto the footholds. However, proper execution of these moves requires practice and a significant amount of strength through the core muscles of the torso. (See chapter 7 to learn exercises for strengthening these core muscles.)

The twist lock is typically used to ease the upgrading of a hand on an overhanging section of wall. For example, consider the situation in which your left hand is on a good hold and you'd like to reach up high with the right hand. While you could attempt this move straight-on—chest facing toward the wall in a neutral position—it's far less strenuous to turn your right hip to the wall before making the reach upward. Proper positioning of the feet is critical for making this move work. Since the right hip is turning to the wall, you'll need to use the outside portion of your right foot on a hold somewhere below or in back of your body (hence the term backstep). Usually you'll find a complementary left foothold to help maintain the twist-lock body position. The feet then press in unison while the left arm pulls down and in toward your torso, creating the twist lock. Finding just the right body position is the key to providing a secure twist lock; when you do, you'll notice that a surprising lack of effort is needed to reach up and acquire the next right handhold. This amazingly efficient locomotion over steep terrain is the magic of the twist-lock technique.

Granted, skillful use of the twist lock and backstep is something that will take many hours of practice. In fact, during your initial attempts at using these moves, you might swear that they require more energy than basic straight-on moves. Trust that with practice you will develop the necessary motor skills to make these moves feel quite easy. Initially limit your practice of the twist lock and backstep to boulder problems that overhang about 20 degrees past vertical. As you acquire skill, expand their use onto even steeper boulder problems as well as overhanging toprope climbs.

Use Creative Footwork with Heel and Toe Hooks and Knee Locks

Heel hooks, toe hooks, and knee locks are real difference makers when it comes to moving over steep terrain and surmounting overhangs. Given the strength of the leg muscles, a good heel hook is often better than a handhold in helping turn the lip of a difficult roof. Similarly, toe hooks and knee locks provide terrific support when you're cranking a serious move on overhanging rock. Let's take a brief look at each.

HEEL HOOK

Heel hooks are the major go-to move when you're turning the lip of a roof, topping out on a boulder



Heel hooks can make a huge difference in pulling a tough roof or copping a shakeout on steep rock.

problem, or copping a quick shakeout while on steep terrain. In these cases you'll likely have your hands on holds at or above face level and intend to place one of your heels on a hold off to the side roughly between hip and shoulder level. Which heel you choose to hook with depends on two factors: the availability of a decent-size hold on which to place your heel, and the location of the next handhold that you hope to acquire. This next reach up is best made with the hand on the same side as the heel hook. So if it looks like the next attainable hold is set up for the right hand, it would be best to use a right heel hook. Once set, pull with the heel hook as if it were a third arm. Often it helps to think about pulling your heel toward your rear end; this will maximize use of the leg muscles and help shift your center of mass toward the heel hook. Meanwhile, the other foot should inside-edge or smear on the wall to help contribute to the upward motion. As you gain elevation, make a quick reach to the next handhold, and then switch your heel hook into a standard step-down foot placement.

TOE HOOK

The toe hook is a foot move used mostly in pulling overhangs or in navigating roofs. This technique involves simply hooking as much of the top/laces side of your shoe as possible on a large protruding hold. Sometimes you will toe hook onto a hold with a bent leg and then straighten that leg as your hands move out the roof. Ideally you'll have one foot toe hooking while the other foot pushes off a nearby hold. This opposing push-pull combination enhances the foot purchase on the holds and lowers the chance that your feet will come swinging off the roof (a common problem). Strive to keep your arms and legs in the straight position as much as possible so that your body weight is being supported more by bones than by muscles. Done properly, you can navigate a surprisingly large roof, with the limiting factor being forearm endurance. Experiment with this foot technique in the bouldering

area and you'll gradually gain skill and confidence in climbing with your back to the ground!

KNEE LOCK (AKA KNEE BAR)

Knee locks are a boon on overhanging walls and roofs with large protruding holds—that is, if you know the technique and can find a position to exploit this “thank-God” move. On a severely overhanging climb with no obvious rest positions, finding a knee lock may be your only hope for a rest. Look for a place where you can step your toe onto a hold and then lock in your knee against a larger opposing hold. Such a knee lock provides surprising purchase; it will allow you to drop one hand at a time to shake out and chalk up. Occasionally you will come upon a knee lock that's so solid, you'll be able to cop a rare no-hands inverted rest! Of course, miss the knee lock and you have no choice but to sprint up the climb in the hope of reaching the anchors before the pump clock runs out.

Use Dynamic Moves When They Offer Greater Economy

Throughout this chapter I've stressed the importance of climbing with maximum economy—the goal being to climb a move, a sequence, and an entire route in a way that requires minimal energy. In most cases the hallmark of economical climbing is smooth, relaxed movement that utilizes the feet and legs over the muscles of the upper body. Such controlled, fluid movement is often referred to as static climbing. The opposite of static style is dynamic or explosive movement, and there are certain moves and sequences that demand dynamic movement to achieve maximum economy. In particular, vertical routes with tiny handholds, and overhanging routes with long reaches often require dynamic movements.

The key is to know which moves are best attempted dynamically versus statically, and this is a recognition skill that takes years to fully develop. When working a route, it's best to attempt a crux

in both a static and dynamic way to determine which style yields the most economical passage; otherwise you'll never know if there was a more effective method, and you'll miss out on an important learning opportunity. Let's examine the two primary forms of dynamic movement: deadpoints and lunges.

DEADPOINTS

Consider a situation in which both hands cling to poor holds, and you would fall off the wall if either hand let go for more than an instant. It's in just such a predicament that the deadpoint move will save the day, because it allows you to make a rapid hand upgrade despite the fact you can't hang on to the wall statically with a single-hand contact point. How's this possible? It's the magic of the deadpoint!

Envision a basketball player making a jump shot. He jumps straight up and shoots the ball at the peak of his flight, a moment of apparent weightlessness before gravity returns him to the floor. This instant of weightlessness and stillness is the deadpoint. Climbers can similarly exploit the apparent weightlessness of the deadpoint to upgrade a hand position as in the desperate situation described above. But instead of jumping like our basketball player, the climber needs to use a smaller, more controlled motion to facilitate the delicate upgrading of a hand from one small hold to another.

For example, imagine a tenuous move on a vertical or slightly overhanging wall in which you want to upgrade your right hand, but you can't make a static reach for the hold. Initiate the deadpoint movement with a small droop downward (or a release outward in the case of an overhanging wall) immediately followed by a firm drawing-inward of the handholds toward your torso. This drawing-in of your body is akin to—but less dramatic than—the basketball player's jump, and there will be an instant when the motion peaks and you'll be able to flash your hand up to snag the next hold. A well-executed deadpoint is calculated and controlled

such that it flows naturally in perfect economy. In extreme cases you may need to execute several deadpoint moves in a row in order to climb through a series of small handholds that you could never hang on to for a static movement.

LUNGES

Unlike the careful, controlled movement of a deadpoint, the lunge (or dyno, as it's often called) is a full-on leap for an out-of-reach hold. In lunging, the arms and legs explode in unison to propel your body upward toward the next good hold. Lunges typically end in one of two ways: Ideally you latch on to a hold and regain control of your body; however, it's also possible that you will fail to catch the target hold and end up falling on the rope or the bouldering crash pad.

Lunging is like any other skill in that it takes practice and a high level of confidence before you will be able to exploit the move in severe situations. It's also a strenuous and stressful move that has led to many shoulder injuries. Consequently, it's best to view lunging as a last-resort move that you only pull out of the bag when nothing else appears possible. In the heat of a crux sequence, though, it often comes down to a gut feeling as to whether you should try to throw a lunge or attempt a static sequence. Ideally it would be best to lunge only when climbing statically would require more energy. In fact, a perfectly executed lunge in just the right situation is a classic example of climbing with high economy, despite the apparent burly nature of the move.

Executing a lunge is very physical, but also requires good timing and a belief that you can reach the next hold. Much like a gymnast attempting her hardest move, throwing and sticking the perfect lunge requires laserlike focus and an intense belief in a successful outcome. Begin by locking your eyes on the target hold and visualize exactly how your hand will hit—and stick!—the hold. Next, look down and concentrate on maximizing an explosive

launch off your four points of contact. In many cases it helps to “cock” your lunge by drooping or bouncing before you catapult upward. As you go airborne, your eyes will naturally return to a pin-point focus on the target hold. Now stick it!

Crack Climbing

Crack climbing involves techniques and tactics far different from the skills outlined above, and therefore you need to develop completely novel motor programs and schema-rules. I advise novice climbers to consult Lisa Gnade and Steve Petro’s *Crack Climbing!* for more comprehensive instruction than the primer I offer here.



Crack climbing is vastly different from face climbing, yet one fundamental remains the same: Let the legs do most of the work. Pumping laps on gym cracks is a great way to hone your technique.

FINGER- AND HAND-JAMMING TECHNIQUES

Just how your hands engage a crack depends on the size of the fissure. The narrowest cracks will accept little more than the tips of your fingers. Most often you’ll place this jam with your index finger on the bottom and your elbow out to the side. This way, when you pull on that arm, the elbow will rotate downward and produce a twisting of the fingers that further anchors them into the crack. One thing you will notice in larger finger cracks (around an inch wide) is that your fingers tend to slide down instead of jamming solidly. In these situations it’s vital to look for constrictions or bottlenecks along the crack and attempt to place your jams in or just above these narrower spots.

Cracks ranging from one to three inches wide are the domain of hand jams. You can execute a hand jam in the thumb-up or thumb-down position. Vertical cracks are often better climbed thumb-down, whereas cracks that slant to the side may be more easily climbed with the lower hand jamming thumb-up and the high hand jamming thumb-down. As in finger jamming, scan hand cracks for constrictions or bottlenecks that will provide the most bombproof jams imaginable. Cracks with little variance are more challenging, as they require a bit more effort to create a solid jam. The key is to cup your hand inside the crack to generate outward pressure and friction on the inside of the crack. Furthermore, jamming thumb-down exerts a twisting force on the jammed hand as you pull down, and this tends to increase the security of the jam.

Fist jamming is the optimal technique for cracks about four inches wide. The technique here is simply to insert your hand with the palm facing into the crack and then make a fist. In closing your hand tightly, the width of your fist increases, making it stick like an oversize cork stuck in the top of a wine bottle. Cracks larger than four inches require a difficult off-width technique in which your arm and leg are jammed to gain purchase.

FOOT TECHNIQUES

There are two primary foot techniques used in crack climbing: feet inside the crack and feet on the rock face. In climbing thin finger cracks, you have little choice but to search for edges on the rock surface on which you can edge or smear (as in face climbing). The exceptions are offset cracks (where one edge of the crack is set out from the opposite edge of the crack) and larger finger cracks, which occasionally afford a foot smear on the exposed edge of the crack. Look for stems, high steps, outside edges, and even backsteps in attempting to use your feet optimally.

Cracks wider than about one inch provide exceedingly solid foot placements by means of foot jamming. The simple technique involves turning your foot sideways—so that the sole of the shoe is facing inward and your knee is bent outward—and inserting as much of your foot as possible into the crack. Depending on the crack's size, you may be limited to jamming just the toe portion of the shoe (narrow hand cracks) or its entire front half (fist cracks). Once you're secured in the crack, your knee will naturally rotate back to center as you stand up on the foot. In climbing a continuous hand or fist crack, you will simply need to leapfrog one foot above the other in a series of foot jams about one foot apart.

LIEBACK TECHNIQUE

Finally, there's liebacking, a unique method of climbing cracks that are located in the vertex of a dihedral or corner. The lieback technique is fairly strenuous since it places your body in a sort of rowing position with your arms pulling and your legs pushing in a powerful opposition. Feet smear on one wall of the corner while your fingers cling to the edge of the crack and arms remain as straight as possible. Move upward by simply walking your feet up the wall and sliding (or leapfrogging) your hands up the crack. With practice you'll learn to position your hands and feet to provide the greatest leverage

at the lowest possible energy expenditure. Of course, as in all kinds of strenuous climbing, moving quickly is fundamental to maximizing performance.

CRACK-CLIMBING STRATEGY

Effective crack-climbing strategy is nearly identical to optimal face-climbing strategy—climb briskly and efficiently, utilize your legs as much as possible in generating locomotion, and pause for long periods only at definitive rest stances. Just as in your clumsy first days attempting any new climbing technique, expect your initial forays at crack climbing to feel awkward and frustratingly difficult. Trust that you will rapidly acquire the unique motor skills and that crack climbing will soon become less strenuous and a heck of a lot of fun.

In your formative days of crack climbing, use the security of a toprope to experiment with different techniques and subtle variations of the basic jamming skills described above. Don't be satisfied with just thrashing up a crack to the top; instead strive to learn the best way to do each move with the goal of climbing each crack in good style and with minimal energy burn. As a practice method, climb single-pitch crack routes a few times in a row to refine your skills and learn the subtle finesse moves that will eventually make you a master crack climber!

Smart Training Drills for Enhanced Learning of Motor Skills and Strategy

Having performed the technical evaluation in table 4.1 and the self-assessment in chapter 2, you should now have a clear picture of your technical weaknesses. But knowledge is not enough; to improve you must act! Dedicate a portion of two climbing sessions per week to improving your weaknesses and you will become a better climber.

Regardless of whether you plan to climb in a gym or at the crag, it's important that you make the distinction between time spent practicing versus

performing. Unfortunately, many climbers handicap themselves by constantly focusing on performance as they succumb to the natural tendency to climb as close to their limit as possible. We've all seen climbers with horrible footwork flailing repeatedly on a steep route so that they can eventually tick some impressively difficult route. While they may ultimately succeed at sending their "5.hard" climbs, they gain little in technical ability from this process—and in fact they further groove their bad habits and poor technique. I call such practices "stupid training."

Conversely, intelligent climbers will dedicate a block of practice time early in the workout when they're mentally and physically fresh. During this practice period or day (if at the crags), the goal is to seek out routes that will target their technical weaknesses. Since performance is not a goal, there is no hesitation to hang on the rope and experiment with moves, body positions, and sequences that feel awkward or difficult.

For example, if backstep and drop-knee moves are a weakness, you'll want to dedicate some time during each workout to getting onto steep routes that demand these moves. No matter your weaknesses, it's likely that the type of routes you must get on will be intimidating, since up to this point it's probably been your tendency to climb routes that favor your strengths. If you are poor at drop-knee moves, for instance, I'll bet you avoid steep routes because they feel especially hard and look overly intimidating to you. The same goes for any style of climbing—slab, thin face, cracks, roofs, what have you. You must partake in regular practice on the terrain and type of routes that target your top technical weaknesses as identified in the self-assessment and table 4.1. Excellence comes no other way.

Following are a few practice drills and games that will enhance your rate of learning new skills and correcting weaknesses, as well as make the process a little more fun. More examples of such speed learning practices can be found in my book *How to Climb* 5.12.



BOULDERING

Bouldering is often touted as the supreme method for developing sport-specific strength, but it's even more effective for learning climbing skills (motor programs) and expanding schema rules. For proof of this, consider a recent bouldering session in which you worked a difficult problem a few times before eventually succeeding. Did you ultimately succeed because your strength increased after each attempt, or did each successive attempt result in learning of the body positioning, feel, and hand- and footwork necessary to do the moves most efficiently and successfully? I think the answer is obvious.

With all the restraints of roped climbing removed, bouldering allows you to narrow your focus and partake in relaxed, repeated attempts at learning a specific skill or sequence of moves. Sports scientists call this blocked practice, because the fixed moves can be practiced over and over again until they are successfully acquired. Once a skill is perfected, however, there is little benefit to additional blocked practice of that skill. Further learning demands that you either move on to practicing a new skill via blocked practice (say, a new boulder problem with new moves and positions) or modify the original problem so that some element of it has changed (say, angle, hold size, hold position or spacing, or the like). This latter strategy is known as variable practice, and it's the gold standard for learning a skill that must be performed in a variety of positions or settings: hitting a golf ball from an infinite variety of lies, shooting a basketball from anywhere on the court, floating a deadpoint from any one of a million different body positions.

Indoor walls and home gyms are the ideal setting for variable practice. Suppose you want to gain skill at, say, using undercling holds and hip turns on overhanging cliffs. To begin, set a problem with a relatively easy sequence of underclings and hip turns. Practice the sequence several times until you feel it's 100 percent wired. Now redesign the

problem with slight changes in the hold positions and locations, and repeat the practice drill until this, too, is wired. Next, reduce the hand- and foothold size and repeat the drill. Keep repeating this process until you've exhausted the possibilities.

Completion of this variable practice drill might take anywhere from a single evening to a couple of weeks. Regardless, the end result is comprehensive schemas surrounding this type of movement, and rapid recall and execution of the skill in some future performance setting. So, while bouldering outdoors on a wide range of move types and angles is best for building a diverse library of climbing skills, using the variable practice strategy on an artificial wall enables comprehensive learning of a new type of movement in a wide range of configurations. Clearly, there is great value in both formats, so get busy!



TRAVERSE TRAINING

Like bouldering, traverse training is a no-frills activity that affords focused practice on numerous technical aspects of the climbing game. Although some people find ad-lib traversing along a cliff base or at a climbing gym boring, this drill does have some major benefits when compared with working a known, graded boulder problem. When working the graded boulder problem, it's natural to want to succeed at any cost, even if your technique is sloppy and inefficient. As discussed earlier, it's difficult to develop new skills in such a performance setting.

Conversely, traversing for the sake of practicing technique and movement eliminates the pressure to perform. You can experiment with new grip positions, gentle and precise foot placements, and various body positions with no concern about whether or not you step off the wall. Maximize the benefit of this drill by carefully spotting each foot placement, concentrating on shifting your center of mass over the leading foot, relaxing your grip as much as possible, and learning to move quickly and confidently through thin, tenuous sequences. Finally, strive to remain calm and relaxed at all times, and

refocus on your feet anytime you sense you're losing control.

To mix things up and increase the intensity and benefit of traverse training, you can also play around with various elimination or focus drills. For example, try doing a complete traverse using only two fingers (the index and middle fingers, for example, or middle and ring fingers) of each hand. This drill forces you to maximize the weighting of your feet (a good thing); it's also an excellent way to increase your finger strength. As another variation, challenge yourself to do a complete traverse using only open-hand finger positions. (This will be especially difficult and beneficial if you naturally favor the crimp grip.) Be creative and make up other drills, such as "side-pull only," "undercling only," or "cross-through only" elimination traverses. Beginner and intermediate climbers have much to gain from performing these drills on a regular basis.



TOPROPING AND HANGDOGGING

Toproping and hangdogging are the ideal formats for practicing difficult moves near your limit or when diving into unfamiliar terrain like pocket or crack climbing. As discussed earlier, a relaxed, low-stress environment is critical to rapid learning of new skills. Obviously, climbing on toprope or on lead, bolt-to-bolt, represents a low-stress setting where you can experiment with tricky, awkward moves without the risk of a serious fall and injury.

When attempting a route that is continuously hard or with multiple cruxes, it's best to break it down into smaller sections or chunks. This reduces the mental burden by allowing you to view and solve the route in parts. Much like working a boulder problem, you can employ blocked practice to work a sequence repeatedly. Once a problem is solved and programmed to a high likelihood of success, you can move on and begin work on the next chunk. Upon solving all the chunks (and after a good rest), your next goal should be to combine chunks. For example, on a route that you had

broken down into four hard sections, you would try to link the top two chunks, and then move down and link the top three chunks. Such incremental learning will wire you for a successful redpoint or top rope ascent after another rest or on your next outing or day at that area. (Note: See chapter 5 of *How to Climb* 5.12 to learn many more strategies for working and sending project routes.)



THE STICK GAME (AKA SEND ME)

This popular game is great for learning to quickly assess and execute a novel, unknown move on-sight—a vital skill when you're on-sight climbing at the crag or in competition. Best played on an indoor wall, the drill requires at least two players who take turns pointing out (with a broomstick or laser pointer) impromptu boulder problems for the other to ascend. Begin by identifying the starting hand- and footholds for the climber; then, as she pulls up on them, the course setter points to the next hold to be used. Continue in this fashion until the climber falls or the problem is done. Commonly, the game is played with open feet—that is, the climber can use any foothold she likes.



FIRST TOUCH

First Touch is a great practice drill for would-be competition and on-sight climbers—though anyone can benefit from its use. An indoor facility with a wide range of top rope routes is the ideal setting for this drill. As the name implies, you must climb a route by using each handhold in the exact way that you first grab it—no readjusting or changing your grip after you first touch it. By climbing many routes in this fashion (and, of course, obeying the guidelines completely), you will learn to examine holds more closely—both from the ground and while climbing—and thus increase the likelihood you'll use them optimally from the first touch. On lead, this skill saves you time and energy, both of which increase your odds of on-sighting a route.



TRACKING AND ELIMINATION

This drill can be used indoors when you're bouldering or on top rope. The goal is simply to climb a route by tracking your feet on the exact same holds used by your hands. It's kind of like climbing a ladder, where you press down on a rung with your right hand and make room to step on that same rung with your right foot. As a type of elimination drill, this will make a climb much harder than it would be if climbed with all the holds on the route. Therefore, if you normally practice skill by climbing 5.10 or 5.11 routes, you'll want to do this drill on routes in the range of 5.8 to 5.9.

If you climb indoors a lot, exercise your creativity and develop other elimination drills that might improve your skill, strategy, and strength. For instance, begin eliminating certain hand- and footholds from the routes you have ruthlessly wired. There's little to be gained from blocked practice of the same tired routes over and over again. So challenge yourself by eliminating the five biggest holds from the route, or by limiting yourself to grabbing the holds only as side pulls or underclings, or with only two fingers, or what have you. Not only can this make for some good fun for you and your friends, but it's also an excellent way to enhance your skill practice and overall ability.



DOWNCLIMBING ROUTES

When I'm leading or top roping indoors, it's rare that I climb a route to the top and lower off without trying to downclimb as much of the route as possible. There are many benefits to this practice beyond the obvious one of doubling the pump. If you know you are going to downclimb a route, you become a more observant and focused climber on the way up. What's more, since poor footwork is a leading handicap for many climbers, there's a lot to be gained from this practice, which demands intense concentration on footwork!

At first you will find downclimbing to be difficult, awkward, and very pumpy. But that's the MO when first attempting anything new that's worthwhile (read challenging). As your hold recognition improves, however, and as you learn to relax and fluidly reverse the route, you'll find that downclimbing a route often feels easier than sending it in the first place. This is because your eccentric (lowering) strength is greater than your concentric (pulling) strength, and due to the fact that by leading with the feet while downclimbing, you learn to maximally weight them and conserve energy. All of the above make downclimbing a killer drill—one not to be overlooked by any serious climber!



SPEED TRAINING

When the rock gets steep and the moves hard, there's no more important weapon to have in your arsenal than being able to climb fast and precisely. Climbing quickly is primarily a function of skill, not strength or power (I'm not talking about lunging wildly up a route). In fact, the less strength and endurance you possess, the more important this skill becomes.

To begin with, it's important to note that there is no benefit to climbing faster if you begin to botch sequences or if your technique degrades. Therefore, you want to practice speed climbing on routes you have completely wired and, likely, at a number grade or two below your personal best. Climb several laps on the route (rest between attempts), each incrementally faster than the previous. Attempt to climb about 10 percent faster on each successive lap, but back off the accelerator at the first sign that your technique is suffering.

Perform this drill a few times a week for several months and you'll find yourself naturally moving faster when climbing on-sight or redpoint at the crags. This new skill alone could push your redpoint ability a full grade higher over the course of a single season—a much greater gain than you'd achieve from strength training alone!

Tips for Developing Superior Technique and Movement Skills

1. Hire a climbing coach. Obtain the objective evaluation and sage guidance of a climbing coach to get on the fast track to climbing excellence. Even just one session per month will be extremely beneficial.
2. Warm up properly before every climbing session. A progressive warm-up reduces injury risk and modulates your neuromuscular system for optimal learning and performance. Spend twenty to thirty minutes on easy climbs and various warm-ups.
3. Embrace a practice perspective and resolve to always have fun. Optimize learning of new skills by maintaining a curious, carefree disposition and having fun regardless of climbing outcomes.
4. Make it your mission to develop a high level of self-awareness. Tune into your body's proprioceptive feedback and learn to really "feel" each move, including the quality of your contact with the holds, the location of your center of mass, and the various levels of tension or relaxation in your muscles. Use this information to discover the most efficient way to do each sequence.
5. Regularly experiment with new techniques and on new terrain. Challenge yourself on a wide range of wall features to expand your skill set and develop confidence. Avoid always climbing known routes and training in the same ways.
6. Push yourself, but know when to call it a day. Spend about one-third of your climbing time on routes just beyond your limit to effectively stretch your boundaries. However, focus the majority of your practice on routes just below your limit to fortify skills and confidence.
7. Get on a regular climbing schedule. Climb two to five days per week (never more) to maximize learning and improvement. Only disciplined, regular practice over the long term will yield consistent improvement and climbing excellence.

*Cameron Horst making the nearly
five-foot jump to the finishing hold
on Raubritze (5.14b), Frankenjura,
Germany. © STEPHAN HORST*

